UTSAV CHAUDHARY

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Computer Science graduate building and running **Python** applications and **scraping bots** for live **data** needs: **Selenium** automation, **LLM**-powered data fetching, **cron**-scheduled jobs, **MongoDB** data cleaning and **streaming pipelines**, and day-to-day **Linux/SSH** operations across **AWS** VMs and **Docker** microservices.

Technical Skills

Scraping, Bots & AI: Python, Selenium, web scraping, data extraction, LLM APIs, AI data-fetching bots, bot automation, REST API testing, FastAPI

Linux, Cloud & DevOps: Linux/Unix, SSH, cron jobs, systemd, AWS, EC2, VMs, Docker, microservices, nginx, log analysis, command-line debugging

Data, DBs & Pipelines: MongoDB, SQL, data cleaning, data sanitization, deduplication, data streaming pipelines, data validation, Pandas

Tools & Practices: Git/GitHub, PM2, JIRA, OpenAI models, Ollama, Redis, monitoring and alerting, CI/CD basics, Agile, remote support, customer-site travel, nights and weekends coverage

Experience

Software Engineer, Hybrid, **Labs196** Feb 2025 - Present

- Developed and maintained **Python** applications and **Selenium** scraping bots for **data fetching**, and integrated **LLM** APIs to classify and summarize fetched content into structured records.
- Deployed and managed all services as **Docker** microservices on **AWS EC2** Linux VMs with **Redis** for **caching**, and set up **cron** jobs and **systemd** timers to start scrapers, sync jobs, and backups on schedule.
- Owned the **MongoDB** data layer and built **data pipelines** in Python to **clean**, **sanitize**, and deduplicate scraped data, then streamed results into downstream collections, dashboards, and APIs.
- Debugged and tuned bots and jobs daily over **SSH** on **Linux**, adding logging, retries, and health checks so failed scrapers and jobs recover automatically.
- Tracked issues and releases through **JIRA** ticket management, monitored bots and jobs from **logs** and alerts daily, and kept **runbooks** updated so any known failure had a logged fix.

Implementation Support Engineer, On-Site, **Midh Technologies**, TX Jan 2024 - Jan 2025

- Supported on-site **software deployment** and customer **go-live** at field operations sites, executing **SIT** and **UAT** test cases and validating **REST API** request/response behavior ahead of **AKS** promotion, maintaining **99.9%** availability.
- Analyzed application and **telemetry** logs on **Linux** via **SSH** to troubleshoot integration issues, cutting mean time to resolution from **6 hrs** to **2 hrs** and documenting each **root cause** in **JIRA**.
- Modified and ran **Python** test and automation scripts, validated operational data with **SQL** queries against **Azure SQL**, and kept **Git** versioning on scripts, configs, and test data.

Projects

Self-Hosted Private Cloud Personal Project

- Designed a private cloud on a custom server rack and replaced paid streaming and cloud storage with self-hosted **Jellyfin**, **Immich**, and **Nextcloud** for media, photos, and files on the home network.
- Wrote **Infrastructure-as-Script** pipelines (**deploy.sh** and per-service scripts) for **Docker Compose**, **PM2**, **nginx**, and **Certbot** so one command deploys the stack with **HTTPS**, and again at boot with **systemd**.
- Run the host on **Arch Linux** with an **Intel Core i7-14700HX**, **32 GB** RAM, an **NVIDIA RTX 4070**, and a **1 TB** NVMe, and keep service data on a **2 TB** disk mirrored daily at 4 AM with an **rsync systemd** timer.
- Built a **FastAPI** dashboard on **PM2** for **10+** projects, with process, **Docker**, port, and **HTTPS** checks plus one-click restart, and run a **kind** lab for Deployments, Services, Secrets, and Ingress.

Automated Ambu Pump (Open-Source COVID-19 Response)

- Built an **automated Ambu pump** with an open-source team during **COVID-19** for **hands-free ventilation**.
- Designed the mechanical drive with **3D-printed** gears and enclosure, **DC motor** and **servo** actuation, and tuned the squeeze cycle for a steady breaths-per-minute rate through hours of continuous running.
- Wrote the **Arduino** control code in **embedded C++** for adjustable rate and depth, and validated squeeze pressure and timing on the bench with the team before each unit was shipped.

Education

State University of New York at Binghamton Aug 2022 to Dec 2023
Master of Science in Computer Science 3.4/4.00 GPA